SERAT BIN AFTAB

AI/ML Engineer | Python Developer | Deep Learning & Computer Vision

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PROFESSIONAL SUMMARY

Computer Science and Engineering undergraduate specializing in Python-based machine learning, deep learning and computer vision. Built and evaluated multi-label retinal-image classifiers (CNN transfer learning, multi-model fusion) and NLP classification pipelines (zero-shot labeling, TF-IDF, ensembles), using patient-level evaluation, threshold tuning and confidence intervals. Seeking AI/ML engineering internships and entry-level roles.

EDUCATION

East Delta University, Chattogram, Bangladesh Expected graduation: February 2027

BSc in Computer Science and Engineering | CGPA: 3.60/4.00 | Thesis defense: November 2026

ACADEMIC PROJECTS

Ocular Disease Prediction Using Deep Learning github.com/SeratAftab/Ocular-Disease-Prediction

Undergraduate Thesis (Team Project) | Primary technical contributor: coding and model training 2026 – Present

Technologies: Python, TensorFlow/Keras, scikit-learn, Pandas, NumPy | Dataset: ODIR-5K (8 disease categories)

- Implemented a multi-label retinal fundus classifier, fine-tuning MobileNetV2, EfficientNetB0, DenseNet121 and ResNet50 with two-phase transfer learning and class-weighted loss to address class imbalance.
- Engineered a four-backbone feature-fusion network with disease-specific prediction heads, and ran a staged ablation (single CNN, concatenation, compression, fine-tuning, loss functions) to isolate each design choice.
- Optimized per-disease decision thresholds; the final model reached macro AUC 0.824 and macro F1 0.591 on the validation split (n = 1,279), compared with 0.815 and 0.567 for EfficientNetB0 alone.

Irreversible Vision Loss Screening github.com/SeratAftab/Irreversible-Vision-Loss

Independent Academic Project | Binary screening for glaucoma, AMD and pathological myopia 2026

Technologies: Python, TensorFlow/Keras, scikit-learn, NumPy, Pandas

- Developed a risk-screening pipeline that aggregates per-disease probabilities from four CNNs and a fusion model using max and noisy-OR rules, with a stacked logistic-regression variant for comparison.
- Evaluated on a patient-disjoint hold-out of 319 images (62 positives), with F1-optimal thresholds tuned on a separate set; the fusion model scored 86.5% accuracy, F1 0.645 and ROC-AUC 0.879 (majority-class baseline: 80.6% accuracy).
- Quantified uncertainty with 2,000-resample patient-level bootstrap intervals (F1 95% CI 0.52–0.74) and compared against a patient-overlapping split to measure leakage-driven optimism.

Healthcare Domain Classification github.com/SeratAftab/Domain-Classification

Independent Academic Project | NLP on HealthCareMagic-100k patient questions 2026

Technologies: Python, scikit-learn, Hugging Face Transformers, XGBoost, Pandas

- Generated domain labels for ~100K patient questions across 8 clinical domains with zero-shot classification (DeBERTa-v3), keeping 85,794 samples at confidence ≥ 0.5 for supervised training on a stratified 70/15/15 split.
- Benchmarked six TF-IDF classifiers (Logistic Regression, linear SVM, XGBoost, Random Forest, Naive Bayes, KNN); raised validation macro F1 from 0.752 to 0.772 via grid search, boilerplate removal and confidence-weighted training.
- Built a calibrated soft-voting ensemble (Logistic Regression + linear SVM): 81.4% accuracy, 0.763 macro F1 and 0.976 macro ROC-AUC on a held-out test set (n = 12,870), measured against the zero-shot-generated labels.

TECHNICAL SKILLS

Languages: Python (advanced), C++ (intermediate), Java (basic), SQL (basic), JavaScript (basic), HTML/CSS
ML / Deep Learning: TensorFlow, Keras, scikit-learn, PyTorch, Hugging Face Transformers, OpenCV, XGBoost
Data & Tools: Pandas, NumPy, Git, GitHub, Bash, Google Colab
Fundamentals: DSA (intermediate), OOP, DBMS, Operating Systems, Computer Networks, Software Engineering

ADDITIONAL INFORMATION

Languages: English (advanced), Bengali (native) | **Open to:** internships and entry-level roles (Bangladesh, remote, abroad)